

DUM IN MIRRIE ISLAND, AustraliaWMO: **941160**Lat: **12.635S**Lon: **130.373E**Elev: **4**StdP: **101.27**Time Zone: **9.50 (AUC)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	17.4	3.4	4.8	23.9	5.4	5.6	23.4	8.1	26.4	6.7	26.7	1.6	130	0.358

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	5.7	33.4	25.1	32.9	25.6	32.6	25.7	28.0	31.5	27.6	31.2	27.3	30.9	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	30.2	26.6	22.2	29.9	26.3	21.8	29.7	89.4	31.6	88.1	31.4	86.8	31.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	7.1	6.2	5.5	13.6	35.3	1.3	0.9	12.7	36.0	12.0	36.5	11.3	37.0	10.4	37.6	29.8
(5)				10.0	28.6	1.0	0.5	9.2	28.9	8.7	29.2	8.1	29.4	7.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	28.6	28.5	28.5	28.5	26.8	24.7	24.1	24.7	27.1	28.8	29.3	29.2
	DBStd	2.31	1.26	1.25	1.10	1.14	1.67	2.10	1.72	1.42	1.36	1.12	1.08	1.28
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6341	576	517	572	553	521	440	436	456	512	583	579	596
	CDD18.3	3299	318	284	314	303	263	190	177	197	262	325	329	337
	CDH23.3	37577	3951	3483	3774	3532	2538	1472	1313	1675	2906	4161	4385	4387
	CDH26.7	14947	1605	1377	1476	1452	945	446	329	443	982	1821	2070	2002
Wind	WSAvg	2.6	3.3	3.2	2.6	2.4	2.4	2.5	2.5	2.4	2.5	2.6	2.6	2.8
Precipitation	PrecAvg	1845	434	390	317	92	28	2	2	4	18	68	141	344
	PrecMax	2501	764	1022	574	417	125	12	26	37	84	208	222	646
	PrecMin	1173	154	59	125	4	0	0	0	0	0	5	78	91
	PrecStd	345	152	218	130	97	35	3	5	8	23	59	41	146
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	32.7	33.0	34.5	33.5	32.5	32.2	32.1	33.0	33.5	33.7	33.6
		MCWB	27.2	27.3	26.4	23.4	22.4	21.5	20.3	18.4	21.3	25.0	26.8	27.1
	2%	DB	32.1	32.1	32.4	33.2	32.5	31.2	30.6	30.7	31.7	32.7	33.0	33.0
		MCWB	27.1	27.1	26.7	23.9	22.9	21.4	20.1	19.6	23.6	25.5	26.5	26.9
	5%	DB	31.6	31.6	31.7	32.5	31.6	30.2	29.5	29.7	31.0	32.1	32.6	32.6
		MCWB	27.0	26.9	26.7	24.6	23.1	21.2	20.4	20.7	24.0	25.5	26.3	26.8
	10%	DB	31.1	31.1	31.2	31.8	30.8	29.3	28.5	28.9	30.4	31.6	32.1	32.1
		MCWB	26.8	26.8	26.5	25.1	23.2	20.9	20.6	21.3	24.0	25.4	26.2	26.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.2	28.2	28.3	28.0	26.5	25.5	24.4	24.7	26.2	27.2	27.8	28.2
		MCDB	31.6	31.6	31.5	31.0	30.2	29.3	28.0	28.5	30.5	31.8	32.3	32.3
	2%	WB	27.7	27.7	27.8	27.1	25.9	24.5	23.7	24.1	25.7	26.7	27.3	27.7
		MCDB	31.1	31.0	30.9	30.4	29.6	28.0	27.4	27.8	29.8	31.3	31.8	31.7
	5%	WB	27.3	27.3	27.4	26.6	25.4	23.8	23.1	23.6	25.3	26.3	27.0	27.3
		MCDB	30.7	30.6	30.6	30.0	29.1	27.3	26.9	27.4	29.2	30.8	31.4	31.3
	10%	WB	27.1	27.0	27.0	26.2	25.0	23.2	22.6	23.2	24.9	26.0	26.6	27.1
		MCDB	30.4	30.2	30.2	29.6	28.5	26.8	26.3	27.0	28.9	30.4	31.0	31.0
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.5	5.3	7.2	8.8	10.0	10.1	9.6	7.1	6.0	5.7	5.1
		MCDBR	5.1	5.2	6.1	8.2	9.2	10.2	10.8	10.5	7.7	6.4	6.1	5.8
		MCWBR	2.3	2.4	2.6	2.8	3.5	4.4	4.7	4.5	3.0	2.5	2.5	2.4
	5% WB	MCDBR	4.7	4.8	5.3	6.5	7.3	8.3	8.7	7.5	6.1	5.7	5.7	5.2
		MCWBR	2.4	2.5	2.7	2.6	2.8	3.5	3.9	3.3	2.5	2.4	2.6	2.4
Clear-Sky Solar Irradiance	taub	0.426	0.413	0.399	0.379	0.375	0.356	0.355	0.370	0.431	0.452	0.455	0.419	
	taud	2.487	2.522	2.564	2.597	2.559	2.555	2.556	2.510	2.364	2.326	2.350	2.494	
	Ebn at Noon	919	924	920	907	878	879	890	902	876	878	886	924	
	Edh at Noon	117	112	105	96	95	92	94	103	126	135	133	116	
All-Sky Solar Radiation	RadAvg	5.17	5.39	5.57	5.78	5.37	5.15	5.38	5.98	6.53	6.79	6.47	5.65	
	RadStd	0.52	0.79	0.68	0.43	0.28	0.18	0.13	0.15	0.21	0.24	0.27	0.56	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	+0.24	N/A	+0.07	N/A	N/A	N/A	N/A

Nomenclature: See separate page